Code And Branch Coverage Report

1. Introduction

Code coverage is a software testing metric used to measure how much of the application's source code is executed during automated testing. This report presents the line coverage and branch coverage results obtained for the Hostel Management System using Pytest and Coverage.py.

2. Objectives

- To measure the percentage of source code statements executed during testing.
- To measure the percentage of decision branches executed during testing.
- To identify uncovered statements and branches.
- To evaluate the effectiveness of the existing automated test suite.

3. Testing Environment

Component	Details
Operating System	macOS, Windows
Programming Language	Python 3.13.1
Application Framework	Flask
Testing Framework	Pytest
Coverage Tool	pytest-cov / Coverage.py
Application	Hostel Management System
Test Database	SQLite

4. Coverage Measurement

Code and branch coverage were measured using the pytest-cov plugin.

Command used:

python -m pytest --cov=app --cov-branch --cov-report=term-missing

The command executes the automated test suite, measures statement and branch coverage for the application package, and displays uncovered lines.

5. Code Coverage Results

The following results were obtained during the coverage execution:

Metric	Result
Total Statements	1,425
Missed Statements	222
Covered Statements	1,203
Line Coverage	84.42%

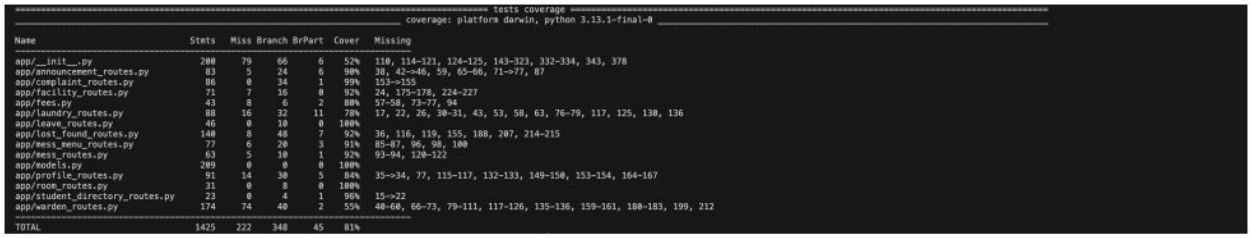
Line coverage indicates that approximately 84.42% of the executable statements in the application package were executed during the automated tests.

6. Branch Coverage Results

Branch coverage measures whether the different decision paths in the application have been executed.

Metric	Result
Total Branches	348
Covered Branches	241
Missing Branches	107
Partially Covered Branches	45
Branch Coverage	69.25%

The results indicate that 69.25% of the measured branches were covered. Some decision paths remain untested and could be addressed through additional test cases.



```
============================= tests coverage ==============================
_____ coverage: platform darwin, python 3.13.1-final-0 _____

Name                                Stmts   Miss Branch BrPart  Cover   Missing
-------------------------------------------------------------------------------
app/__init__.py                       200     79     66      6    52%   110, 114-121, 124-125, 143-323, 332-334, 343, 378
app/announcement_routes.py             83      5     24      6    90%   38, 42->46, 59, 65-66, 71->77, 87
app/complaint_routes.py                86      0     34      1    99%   153->155
app/facility_routes.py                 71      7     16      0    92%   24, 175-178, 224-227
app/fees.py                            43      8      6      2    80%   57-58, 73-77, 94
app/laundry_routes.py                  88     16     32     11    78%   17, 22, 26, 30-31, 43, 53, 58, 63, 76-79, 117, 125, 130, 136
app/leave_routes.py                    46      0     10      0   100%
app/lost_found_routes.py              140      8     48      7    92%   36, 116, 119, 155, 188, 207, 214-215
app/mess_menu_routes.py                77      6     20      3    91%   85-87, 96, 98, 100
app/mess_routes.py                     63      5     10      1    92%   93-94, 120-122
app/models.py                         209      0      0      0   100%
app/profile_routes.py                  91     14     30      5    84%   35->34, 77, 115-117, 132-133, 149-150, 153-154, 164-167
app/room_routes.py                     31      0      8      0   100%
app/student_directory_routes.py        23      0      4      1    96%   15->22
app/warden_routes.py                  174     74     40      2    55%   40-60, 66-73, 79-111, 117-126, 135-136, 159-161, 180-183, 199, 212
-------------------------------------------------------------------------------
TOTAL                                1425    222    348     45    81%
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(.venv) kundanveluru@Kundans-MacBook-Pro SE-Hostel_Management_System % coverage json -o coverage.json && python -c "import json; print(json.load(open('coverage.json'))['totals'])"
Wrote JSON report to coverage.json
{'covered_lines': 1203, 'num_statements': 1425, 'percent_covered': 81.44388042865201, 'percent_covered_display': '81', 'missing_lines': 222, 'excluded_lines': 0, 'percent_statements_covered': 84.42105263157895, 'percent_statements_covered_display': '84', 'num_branches': 348, 'num_partial_branches': 45, 'covered_branches': 241, 'missing_branches': 107, 'percent_branches_covered': 69.25287356321839, 'percent_branches_covered_display': '69'}
```

7. Test Execution Summary

Metric	Result
Total Tests Executed	223
Passed	223
Failed	0
Test Execution Time	48.62 seconds
Combined Coverage	81%

All 223 automated tests passed successfully. The coverage execution also reported warnings, which did not prevent the tests from passing.

8. Analysis

The Hostel Management System achieved 84.42% line coverage and 69.25% branch coverage.

The line coverage result indicates that most of the application's executable statements were exercised by the automated test suite. Branch coverage is comparatively lower, indicating that certain alternative decision paths and conditions require additional testing.

The coverage report also identifies specific source files and line numbers that were not covered, helping identify areas where further test cases can be introduced.

9. Conclusion

Code and branch coverage analysis was successfully performed on the Hostel Management System using Pytest and Coverage.py.

The application achieved 84.42% line coverage and 69.25% branch coverage, with all 223 automated tests passing successfully.

The results demonstrate that the existing automated test suite exercises a substantial portion of the application. Additional test cases can further improve branch coverage and test the remaining uncovered execution paths.